

04 — Technical documentation

PauseAm — Ask before you pay. Innovation & Enterprise / PS2. Review build, 2 October 2026.

Implemented behaviour and limits

A no-account mobile interface offers before-payment, after-payment and learning journeys. Typed English situations retrieve source-linked CBN checklists covering supplier changes, school fees, receipt claims, impersonation, banking codes, shopping, investment warnings and suspected fraud. It never declares a person, account or payment safe. Source checks are automated; human safety review is pending.

After-payment bank-contact advice is immediate. Incident notes contain only selected event categories and times in page memory. Downloads and sharing exclude questions, transcripts and credentials. The learning interaction explains a code-sharing trap. User usefulness has not yet been demonstrated.

Four official ASR routes and an N-ATLaS text adapter exist. The real text weights loaded successfully; the first inference failed the output contract and was withheld. Four-language comprehension and fluent review are unverified. Current interface and source wording are English. No official API success, fine-tuning, receipt authentication or screenshot redaction is claimed.

Architecture and data flow

Frontend: React 19, Vinext, Vite, TypeScript, system fonts, responsive CSS and existing Radix controls. Backend: same-origin Cloudflare Worker routes, no bound database. Inference: separately configured Python 3.11 FastAPI HTTPS services.

Voice: consent → record up to 28 seconds → listen/discard → mono 16 kHz PCM16 WAV → bounded upload → authenticated host → exact pinned model → contract checks → user-corrected and confirmed transcript → source-grounded checklist. Browser recognition is not a fallback. Browser read-aloud is labelled and receives public guidance only.

ASR follows the official Transformers pipeline. Current weight files are `pytorch_model.bin`, so the service uses torch 2.6 weights-only loading with remote code disabled. Text uses official Safetensors and the documented chat-template path. The text host authenticated health response confirms the pinned revision, verified bucket provenance and bfloat16 load. ASR weight loading remains blocked on file access.

Setup and reproducibility

Use Node ≥ 22.13 and Python 3.11 for inference. Clone the public repository. Run `npm ci --no-audit --no-fund`, `npm run dev` and open the printed local URL. Run `npm test`, `npm run typecheck`, `python -m unittest tests.test_audio -v`, `npm run build`, and `npm run test:http` against the running server. For production preview run `npm run start` after building.

The JavaScript lockfile is committed. New npm install-script policy allows only six pinned esbuild, workerd and unrs-resolver build dependencies; no blanket script permission is enabled. Temporary verification clones are excluded from TypeScript scanning. The Windows npm.cmd shim may need the installed `npm-cli.js` invoked through node. Python direct dependencies are pinned. Docker base images and build actions are pinned to digests/commits. Actual model images, offline import checks and resolved dependency snapshots are recorded in `submission/evidence/model-container-build-2026-10-02.json` and `submission/evidence/model-container-runtime-2026-10-02`. Use the actual image digests for the reproducible pilot; source rebuilds must reconcile their resolved dependencies. Actual text loading and the first failed inference are recorded in `submission/evidence/text-inference-first-attempt-2026-10-02.json`. This is a synthetic engineering input and a genuine model response, not a participant interaction.

Keep tokens in server-side host secret settings. Never put them in a NEXT_PUBLIC variable, source, screenshot or chat. Use .env.example for key names and safe defaults. .env, .dev.vars, private research, raw recordings and identity documents are ignored. See model-hosting.md for deployment and secure account setup.

API contracts

GET /api/status returns pinned targets, library version and configuration flags. It performs no inference; configuration is not validation.

POST /api/asr requires matching Origin, Content-Type audio/wav, X-Audio-Consent: yes and X-Language en/yo/ha/ig. Body: canonical PCM16 mono 16 kHz WAV, 0.5–30 seconds, maximum 960,044 bytes. Success: actual text, model, revision, language, confidence:null, latencyMs. Errors: invalid consent/language 400, origin 403, media 415, private output 422, capacity 429, inference unavailable 503. No synthetic transcript.

POST /api/answer accepts question (1–600 characters), journey before/after/learn and language. Privacy/source checks precede inference. Urgent response does not await a model. The optional text service constrains actual model token choices to JSON containing at most one trusted card ID or abstention; independent validation follows. Wrong identity, unknown IDs, malformed or oversized responses fail closed.

Model-host /ready reports readiness; authenticated /health returns model/revision. ASR /transcribe accepts WAV. Text /guide accepts the corrected situation and server-supplied source context. These are team-host contracts, not official API proof.

Evaluation JSON schema 2 labels its four-model catalog as targets, not inference evidence. Consented successful ASR trials retain only verified model ID/revision, selected language and anonymous counts/latency. Language rows do not assign old records without a target to English. No text, audio or participant identifier is retained. Local records can be modified by the browser owner and are not audited field validation.

Source management

lib/safety.ts is the versioned registry: URL, section, checked date, expiry and review status accompany each entry. Scenario adaptations are distinguished from CBN facts. Source details are expandable next to answers. The complaint PDF returned 403 during recheck, so the official Complaint Lodgment HTML page is used instead. Users check current applicable complaint timelines at the source, while urgent fraud reporting is immediate.

Proposed owners: Kayode coordinates safety review; Suleiman deploys approved content. Review monthly and after reported changes. This version expires 1 November 2026. Disable uncertain entries. Changes need current source evidence, human sign-off, tests and a version bump. No human sign-off has been invented.

Privacy and security

Do not enter names, account details, PINs, OTPs, passwords or full credentials. The guard rejects digits, emails, common secret phrases and English digit-word sequences. It cannot recognise every sensitive expression or language. Speakers must listen before uploading. Screenshot input stays disabled.

Input text/audio are transient. Mutable PCM buffers are cleared where possible; managed-runtime copies cannot be guaranteed erased immediately. No application database or request-body/transcript log exists. Operators must verify host and proxy logging/retention independently. Network providers may process IP metadata.

Protection includes origin checks, bounded streams, HTTPS-only configured hosts, redirect refusal, pinned identity and strict output allowlists. Answer requests are limited to 90/minute and ASR to 30/minute per isolate; model hosts serialize inference. Add distributed edge limits before public model use; these controls are not a global abuse, spending or licence cap. No automatic inference retry re-uploads audio. Users may retry deliberately after a failure.

Pasted messages are untrusted data. Retrieval cannot execute instructions. Model outputs cannot introduce links, contacts or free-form advice. Relevance and prompt injection robustness still need real-model adversarial evaluation; contract tests do not prove immunity.

Accessibility and performance

Semantic headings, labelled inputs, keyboard controls, skip link, visible focus, large touch targets, high-contrast colours, reduced motion and a text alternative are implemented. Record actual browser, screen-reader and mobile checks separately. No WCAG certification is claimed; TalkBack/NVDA and fluent-language testing remain.

Web Vitals measures real LCP, INP and CLS only in page memory. Values are not uploaded. Record device, viewport, network/CPU conditions, cache, build and sample count. Development cold compilation is not representative mobile performance. Separate ASR/answer latency from page responsiveness. Targets: LCP ≤ 2.5 s, INP ≤ 200 ms, CLS ≤ 0.1 . Targets are not achieved results.

Tests and deployment

Tests cover retrieval, expiry, urgent priority, abstention, privacy, telemetry whitelists, Unicode WER, invalid audio, stream bounds, cancellation, wrong language or model identity, unavailable endpoints and constrained source selection. Transport fixtures are explicitly test-only and excluded from validation. Python checks cover truncated/oversized/wrong-format PCM; HTTP checks exercise actual app routes without claiming model inference. Eighteen Python checks now include the finite token grammar and output boundary. The complete root Igbo model was byte-verified; the mixed bucket needs isolated text/en/yo/ha/ig folders. Final digest-pinned model images built successfully. User-operated deployment and real-host checks are in deployment/secretvm/README.md.

Build the Worker, push its exact commit, package and deploy through Sites. Validate public fresh-session access on desktop/mobile. Configure model hosts separately, verify /health, then run genuine consented WAV requests and the full correction flow in every claimed language. ASR_ENABLED=false and TEXT_ENABLED=false are kill switches. Roll back the previous site version if release checks fail.

Fine-tuning

Not performed. After access, reviewed data and suitable compute, use narrowly scoped card-selection LoRA/QLoRA. Split by scenario/person before training; de-duplicate; freeze multilingual, adversarial and abstention evaluation cases. Use only consented or authored, reviewed non-sensitive data with documented rights. Record hashes, seed, hardware, time, hyperparameters and adapter checkpoint.

Compare unchanged pinned N-ATLaS and the adapter on identical held-out cases. Report selection accuracy, abstention, unsupported advice, per-language human review and latency/cost. Downloading or running weights is not fine-tuning. No dataset, run or improvement has been fabricated.

Sustainability and licence

Local GPU: Quadro P520, 4 GiB. It is not evidence of an 8B training run. The inspected Google Cloud SecretVM quote is \$0.52/hour for eight vCPUs, 32 GB RAM and 160 GB storage, or \$0.052/hour while stopped. A Production pilot was launched on 2 October. Authenticated loading passed; the first actual guidance request failed its contract in 30,453 ms. A later image update reported a provider error; recovery and successful guidance remain pending. Throughput and costs per successful interaction remain unmeasured. Calculate costs from measured host-hours, actual dated rates, egress, storage, monitoring, content review and support; disclose assumptions and currency. No unverified price is a cost result.

Model terms cap use at 1,000 active end users per rolling 30 days. Obtain separate Awarri/FMCIDE licensing before exceeding it. Restrict the pilot and verify cross-host user accounting before public inference. The prepared deployment uses one shared SQLite quota of 950 inference reservations per rolling 30 days, including

failures. This is deliberately stricter than active-person accounting. Never reset or fork the database to bypass the cap. A cookie is not reliable person counting. Required model attribution is in the app and integration register. Public source availability does not grant rights to third-party model weights.

Remaining release gates

Four ASR weight loads, successful guidance traces, official service/API clarification, fluent review, human safety review, 50 documented interactions, representative mobile/assistive testing, real video, review of the provided team facts and private ID/CAC are pending. ONDI pages 1–2 were inspected and local drafts fit verified limits; Programme Fit and remaining constraints are pending. Section navigation auto-saved the pre-existing draft without field edits. The session subsequently returned to sign-in. Nothing has been submitted.